

ADOS-2 | Autism Diagnostic Observation Schedule, Second Edition

Lord, Rutter, DiLavore, Risi, Gotham, & Bishop (2012)
Lord, Luyster, Gotham, & Guthrie (2012)

 unlocking potential

ADOS[®]-2 Training for Clinicians

Philadelphia, PA / Day 2: May 2, 2025
A Live Event by The WPS Professional Learning Center (ProLearn[®])

Michael Morrier, PhD, BCBA-D

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ADOS-2 Administration Modules 1 and 2

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Today's Agenda

- Remaining questions about Modules 3 or 4?
- Introduction to Modules 1 and 2
- Demonstration of Module 1 or 2
- Coding
- ADOS-2 Psychometrics
- Appropriate use of the ADOS-2
- Information for further preparation and training

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Administration Recap Modules 1 and 2

- Make sure the environment is properly arranged
 - Appropriately sized table and chairs
 - Space to set out toys and move around
- Double-check that materials are working
 - E.g., bunny
- Have a parent or familiar caregiver in the room
 - Remember to prep them and ask about food restrictions
- Take a language sample and be ready to switch modules at any time
 - Build in time for a brief warm-up period (especially if ADOS-2 is the only assessment administered that day)

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Guidelines for Selecting Module

Language **FIRST** and **THEN** age

(see ADOS-2 Manual, p. 10)

Module	Expressive Language Level Requirements	Chronological Age Recommendations
T	No speech to simple phrases	12 to 30 months
1	No speech to simple phrases	At least 31 months; may not be appropriate past school age
2	Three-word phrases/ not yet verbally fluent	May not be appropriate past school age
3	Regular use of complex sentences	Child or younger adolescent (≈16 and younger)
4	Regular use of complex sentences	Older adolescent or adult

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Special Administration Notes Modules 1 and 2

- Previous testing experience with very young children is crucial for Modules 1 and 2
 - Young children are supposed to be wary of new situations and people (e.g., excessive animation may overwhelm a young child)
- Don't forget to play! It should be fun!

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DURING the Administration

- Move from less structure to more structure.**
 - What do you have to do to elicit communication of various types?
 - What does it take to establish a successful social interaction?
 - If the child shows an intense interest or repetitive behavior, how difficult is it to move them onto something else?
- Be flexible—with the materials, with the space, with yourself and your behavior.
 - You CAN administer tasks in any order.
- Keep pace with the child.
 - Try to match affect, activity, and language level of child.
- Take notes!**

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Module 1 Activities Pre-Verbal/Single Words

31 Months and Older

- Free Play
- Response to Name
- Response to Joint Attention
- Bubble Play
- Anticipation of a Routine With Objects
- Responsive Social Smile
- Anticipation of a Social Routine
- Functional and Symbolic Imitation
- Birthday Party
- Snack

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Introducing Modules 1 and 2 to Parent/Caregiver

- Inform the parent/caregiver that you will attempt to complete some activities with the child and ask them for help when needed.
- Encourage the parent/caregiver to respond as they normally would if the child approaches them.
- If necessary, remind the parent/caregiver to avoid making comments or directing the child too quickly.
- During Free Play, ask the parent if these are the types of toys the child normally plays with at home.

(see ADOS-2 Manual, p. 27)

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Free Play



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Response to Name

1. Call the participant's name UP TO four times.
 - If the participant makes an appropriate vocal response without making eye contact with you upon one of the first two presses, start the task over with four new tries calling the participant's name.
2. Ask the parent/caregiver to call the participant's name UP TO two times.
3. Ask the parent/caregiver to say other words or make a familiar noise or sound.
4. Ask the parent/caregiver to do whatever is necessary, including touching the participant, to get them to look at the parent/caregiver.

(see ADOS-2 Manual, p. 27)

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Response to Joint Attention

1. Call participant's name and/or touch them to get their attention UP TO FIVE times.
2. Say: "[Participant's name], look!" with exaggerated shift gaze toward toy, then look at participant UP TO THREE times.
3. Say: "[Participant's name], look at that!" with exaggerated shift gaze toward toy, then look at participant UP TO THREE times.
4. Say: "[Participant's name], look at that!" and repeat the exaggerated gaze shift, then point at the toy UP TO TWO times.
5. Activate toy and look for appropriate response.
6. Turn off toy and pause for 5 seconds.
7. Place toy in front of participant.
8. Turn toy on for 5 seconds, then turn it off and wait for reaction.
9. Allow play for appropriate amount of time.



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Bubble Play



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Bubble Play (cont.)

1. Blow bubbles with toy away from your body without calling attention.
2. Continue for 5 seconds after participant notices the bubbles.
 - Note reactions to bubbles.
3. Match/encourage positive affect, and model fun behaviors to obtain engagement if needed.
4. Establish a routine for playing with bubbles.
5. Allow participant to request more bubbles by waiting for their initiation.
6. Play for a few minutes—enjoyable for many participants.

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Anticipation of a Routine With Objects



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Anticipation of a Routine With Objects (cont.)

1. Blow up balloon with exaggerated behavior, then allow participant to hit it.
2. Alert the participant before holding balloon over your head and letting it go.
3. Wait for participant to bring you balloon or indicate they want it again.
 - Indication made: repeat routine.
 - No response: encourage participant to return balloon to you.
 - Participant throws balloon/loses interest: you pick it up and move to step 4.
4. Get balloon, show it to participant, and repeat routine in deliberate steps:
 - Hold balloon in front of your mouth.
 - Put balloon to your mouth.
 - Blow up balloon.
 - Hold inflated balloon over your head.
 - “Ready, set, go!”
 - Release balloon.
5. Repeat a few times, waiting for initiation from participant each time.

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Responsive Social Smile

Evaluate the occurrence of the child's smile in response to:

- a) You smiling
- b) The parent/caregiver smiling
- c) The parent/caregiver smiling and making a familiar noise or calling in a way that implies physical contact but without actually touching the child, or
- d) Being touched

This is a discrete task: decide when to carry out this task, then do so.

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Anticipation of a Social Routine



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Anticipation of a Social Routine (cont.)



- Peekaboo: Hold the blanket up between your face and the child. Say, "Where's [child's name]?" Pull the blanket down with excitement, saying "Peekaboo!" or the child's name. Repeat once, hesitating with the blanket held up. If the child pulls the blanket down, repeat, then pause and hold the blanket out toward the child. Watch for a move to put the blanket on your face, on their face, or to look at you, waiting for you to repeat. If not, try once more, then proceed to tickling and swinging.
- Tickling: Say, "Here comes the tickle bug!" Hold your hands up and move them in a tickling motion while moving toward the child. Gently tickle their leg or stomach.
- Swinging: Approach with arms out and "jump" the child up and down lightly, counting down before each jump. On the second approach, wait before jumping.

*If the child does not respond positively, ask the caregiver to try the same games. If the child still does not respond positively, ask the caregiver to demonstrate a social routine familiar at home.

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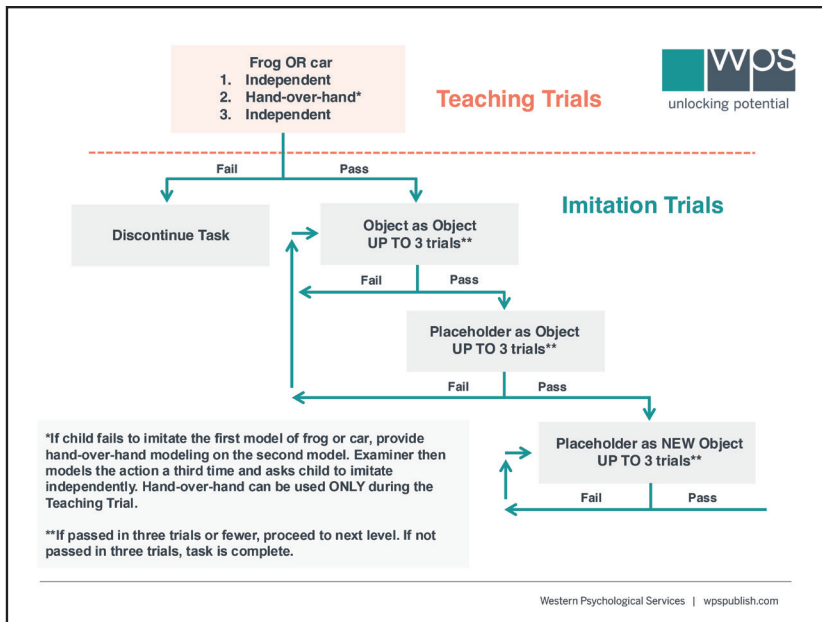
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Functional and Symbolic Imitation

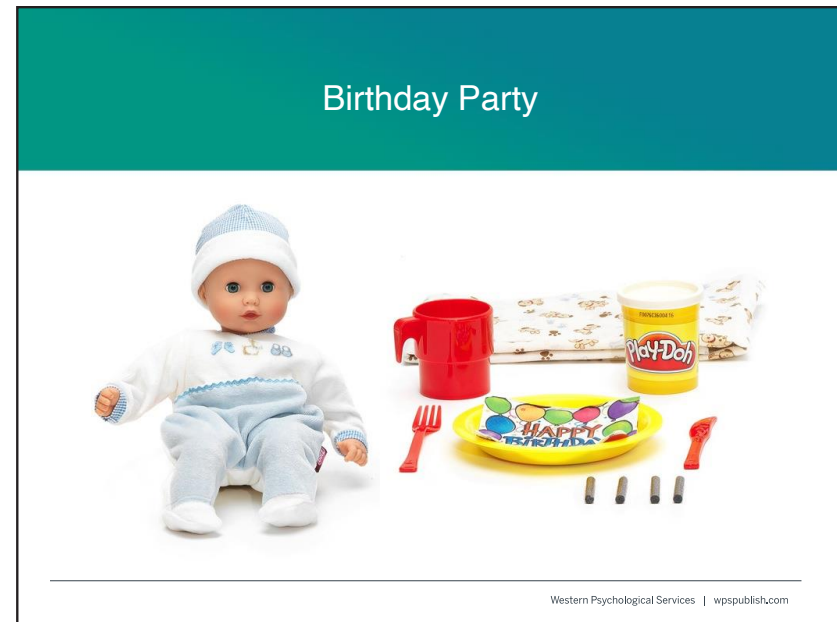


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Birthday Party*

Modules 1 and 2 Administration

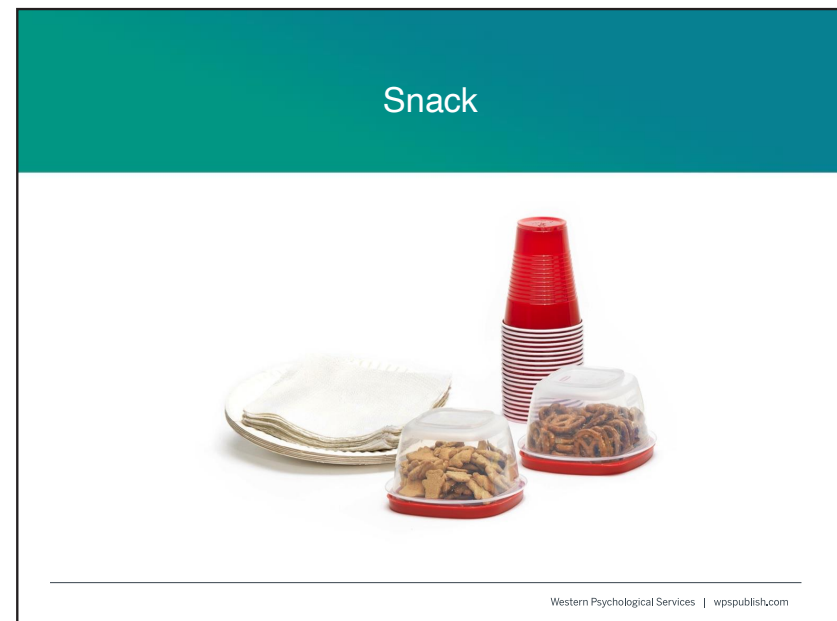
- Begin each step with a basic verbal cue: "What should we do next?"
- Present each step to give all participants the same opportunities.

1. Using the doll, Play-Doh, candles, and utensils, walk the participant through the steps of a birthday party.
2. Introduce birthday cake.
3. Introduce candles and get participant to blow them out with as few cues as possible.
4. Encourage the participant to feed the baby (e.g., say "Baby is hungry!").
5. Pretend to spill juice and have the participant help clean it up.
6. Have the participant put the baby to bed.

**Modifications required if child's family does not celebrate birthdays.*

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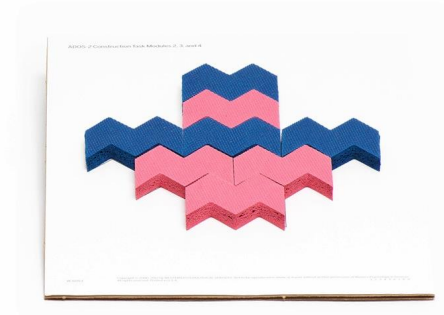
Module 2 Activities (Phrase Speech)

1. Construction Task
2. Response to Name
3. Make-Believe Play
4. Joint Interactive Play
5. Conversation
6. Response to Joint Attention
7. Demonstration Task
8. Description of a Picture
9. Telling a Story From a Book
10. Free Play
11. Birthday Party
12. Snack
13. Anticipation of a Routine With Objects
14. Bubble Play

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Module 2 Administration Construction Task



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Make-Believe Play and Joint Interactive Play



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Conversation

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Demonstration Task



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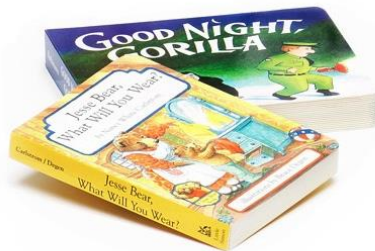
Description of a Picture



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Telling a Story From a Book



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Acceptable Modifications to Modules 1 and 2

- For older youth who have limited language
 - Using action figures in place of baby doll
 - Adding action figures and Break materials from Module 3 to the Free Play
 - Introducing two bubble toys during Bubble Play so that the examiner and participant can have a "bubble fight" (using a single tray of bubble fluid, which the examiner maintains control of)
 - Using two plates and cups during Snack so that the examiner and participant can share a snack together
 - Age-appropriate social routine (high five or hand-slapping game, spin in a chair instead of tickling and swinging in air)
 - Avoiding terms "toys" and "play"
 - Age-appropriate statements in social smile ("Are you having fun?" vs. "Look at that tummy!")
 - **Adapted ADOS (A-ADOS) is available for research use (Bal et al., 2020)**

(see ADOS-2 Manual, p. 14)

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Read Through the Coding Sections

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List of Codes: Module 1

A Codes: Language and Communication	B Codes: Reciprocal Social Interaction	
Overall Level of Non-Echoed Spoken Language	Unusual Eye Contact	Spontaneous Initiation of Joint Attention
Frequency of Spontaneous Vocalization Directed to Others	Responsive Social Smile	Response to Joint Attention
Intonation of Vocalizations or Verbalizations	Facial Expressions Directed to Others	Quality of Social Overtures
Immediate Echolalia	Integration of Gaze and Other Behaviors During Social Overtures	Amount of Social Overtures/Maintenance of Attention: EXAMINER
Stereotyped/Idiosyncratic Use of Words or Phrases	Shared Enjoyment in Interaction	Amount of Social Overtures/Maintenance of Attention: PARENT/CAREGIVER
Use of Another's Body	Response to Name	Quality of Social Response
Pointing	Requesting	Level of Engagement
Gestures	Giving	Overall Quality of Rapport
	Showing	

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List of Codes: Module 1 (cont.)

C Code: Play	D Codes: Stereotyped Behaviors and Restricted Interests	E Codes: Other Abnormal Behaviors
Functional Play With Objects	Unusual Sensory Interest in Play Material/Person	Overactivity
Imagination/Creativity	Hand and Finger and Other Complex Mannerisms	Tantrums, Aggression, Negative or Disruptive Behavior
	Self-Injurious Behavior	Anxiety
	Unusually Repetitive Interests or Stereotyped Behaviors	

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List of Codes: Module 2

A Codes: Language and Communication	B Codes: Reciprocal Social Interaction	
Overall Level of Non-Echoed Spoken Language	Unusual Eye Contact	Quality of Social Overtures
Speech Abnormalities Associated With Autism	Facial Expressions Directed to Others	Amount of Social Overtures/Maintenance of Attention: EXAMINER
Immediate Echolalia	Shared Enjoyment in Interaction	Amount of Social Overtures/Maintenance of Attention: PARENT/CAREGIVER
Stereotyped/Idiosyncratic Use of Words or Phrases	Response to Name	Quality of Social Response
Conversation	Showing	Amount of Reciprocal Social Communication
Pointing	Spontaneous Initiation of Joint Attention	Overall Quality of Rapport
Descriptive, Conventional, Instrumental, or Informational Gestures	Response to Joint Attention	

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List of Codes: Module 2 (cont.)

C Code: Play	D Codes: Stereotyped Behaviors and Restricted Interests	E Codes: Other Abnormal Behavior
Functional Play With Objects	Unusual Sensory Interest in Play Material/Person	Overactivity
Imagination/Creativity	Hand and Finger and Other Complex Mannerisms	Tantrums, Aggression, Negative or Disruptive Behavior
	Self-Injurious Behavior	Anxiety
	Unusually Repetitive Interests or Stereotyped Behaviors	

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ADOS-2 Administration and Independent Scoring

Module 1 or 2

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Algorithm

- For Modules T-2, select appropriate algorithm based on age and/or code on Item A1: Overall Level of Non-Echoed Spoken Language
- Transfer selected scores from protocol to algorithm
- Convert scores
 - 3 to 2
 - 7, 8, and 9 to 0

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Classification: Modules 1 and 2

- Compute algorithm domain total scores for Social Affect (SA) and Restricted and Repetitive Behavior (RRB)
- Combine SA + RRB totals = Overall Total
- Compare the Overall Total to the cutoff scores provided for each algorithm
- This will result in an ADOS-2 *Classification*, NOT a *diagnosis*

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Classification: Module 1

- Compare Overall Total to the cutoff scores

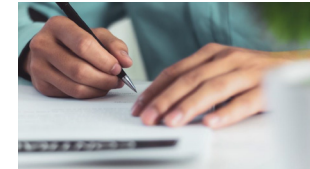
- Fill in the ADOS-2 Classification based on overall total score
 - Classification, NOT a diagnosis

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Comparison Score: Modules 1 and 2

- Use Overall Total score and chronological age to compute Comparison Score
- Comparison Score tables are provided in Manual and on protocols



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Comparison Score: Module 1

- Convert Overall Total Score to Comparison Score

- Find the Comparison Score in the Table

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Domain Level Comparison Score Module 1 Social Affect

Social Affect CS	Module 1, No Words			Module 1, Some Words		
	2 years	3 years	4-14 years	2-3 years	4 years	5-14 years
1	0-3	0-3	0-2	0-1	0-1	0-1
2	4-5	4-5	3-5	2-4	2-3	2-3
3	6-8	6-9	6-9	5	4-5	4-5
4	9	10	10	6-7	6-7	6-7
5	10-13	11-12	11-12	8	8-9	8-9
6	14-16	13-16	13-14	9-11	10-12	10-13
7	17	17	15-16	12-13	13	14-15
8	18	18	17-18	14-15	14-15	16
9	19	19	19	16-17	16-17	17-18
10	20	20	20	18-20	18-20	19-20

Source: Hus, V., Gotham, K., & Lord, C. (2014). Standardizing ADOS domain scores: Separating severity of social affect and restricted and repetitive behaviors. *Journal of Autism and Developmental Disorders*, 44(10), 2400-2412.

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Domain Level Comparison Score Module 1 RRB

RRB CS	Module 1, No Words			Module 1, Some Words		
	2 years	3 years	4–14 years	2–3 years	4 years	5–14 years
1	0	0	0	0	0	0
2						
3						
4						
5	1	1	1–2	1	1	1
6	2	2–3	3	2	2	2–3
7	3	4	4	3	3–4	4
8	4	5	5–6	4	5	5
9	5	6	7	5	6	6
10	6–8	7–8	8	6–8	7–8	7–8

Source: Hus, V., Gotham, K., & Lord, C. (2014). Standardizing ADOS domain scores: Separating severity of social affect and restricted and repetitive behaviors. *Journal of Autism and Developmental Disorders*, 44(10), 2400–2412.

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Classification: Module 2

- Compare YOUR Overall Total to the cutoff scores

- Fill in the ADOS-2 Classification based on overall total score
 - Classification, NOT a diagnosis

Converting the Overall Total to the ADOS-2 Classification and ADOS-2 Comparison Score

ADOS-2 Classification	ADOS-2 Comparison Score
1	0–1
2	2–3
3	4–5
4	6–7
5	8–10
6	11–13
7	14–15
8	16–17
9	18–20

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Comparison Score: Module 2

- Convert Overall Total Score to Comparison Score

- Find the Comparison Score in the Table

ADOS-2 Comparison Score

In the table below, select the column corresponding to the child's chronological age. Next, locate the obtained Overall Total in that column. Follow the row with the obtained Overall Total to the left or the right column to find the ADOS-2 Comparison Score associated with that Overall Total.

Overall Total	2	3	4	5–6	7–8	9–10	11–13	14–15	16–17	18–20
1	0–1	0–1	0–1	0–1	0–1	0–1	0–1	0–1	0–1	0–1
2	2–3	2–3	2–3	2–3	2–3	2–3	2–3	2–3	2–3	2–3
3	4	4	4	4	4	4	4	4	4	4
4	5	5	5	5	5	5	5	5	5	5
5	6	6	6	6	6	6	6	6	6	6
6	7	7	7	7	7	7	7	7	7	7
7	8	8	8	8	8	8	8	8	8	8
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74	75	75	75	75	75	75	75	75	75	75
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76	77	77	77	77	77	77	77	77	77	77
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93	94	94	94	94	94	94	94	94	94	94
94	95	95	95	95	95	95	95	95	95	95
95	96	96	96	96	96	96	96	96	96	96
96	97	97	97	97	97	97	97	97	97	97
97	98	98	98	98	98	98	98	98	98	98
98	99	99	99	99	99	99	99	99	99	99
99	100	100	100	100	100	100	100	100	100	100

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Domain Level Comparison Score Module 2 Social Affect

Social Affect CS	2–3 years	4 years	5–6 years	7–16 years
1	0–1	0–1	0–1	0–1
2	2–3	2	2–3	2
3	4	3–4	4–5	3–4
4	5	5–6	6	5
5	6	7	7	6–7
6	7–8	8–9	8–9	8–10
7	9–10	10–11	10–11	11–13
8	11	12–13	12–15	14–15
9	12–14	14–15	16	16–17
10	15–20	16–20	17–20	18–20

Source: Hus, V., Gotham, K., & Lord, C. (2014). Standardizing ADOS domain scores: Separating severity of social affect and restricted and repetitive behaviors. *Journal of Autism and Developmental Disorders*, 44(10), 2400–2412.

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Domain Level Comparison Score Module 2 RRB

RRB CS	2-3 years	4 years	5-6 years	7-16 years
1	0	0	0	0
2				
3				
4				
5	1	1	1	1
6	2	2-3	2-3	2-3
7	3	4	4	4
8	4	5	5	5
9	5-6	6	6	6
10	7-8	7-8	7-8	7-8

Source: Hus, V., Gotham, K., & Lord, C. (2014). Standardizing ADOS domain scores: Separating severity of social affect and restricted and repetitive behaviors. *Journal of Autism and Developmental Disorders*, 44(10), 2400-2412.

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Calculating Reliability

- Goal: **At least 80%** interrater agreement or agreement with consensus on overall protocol and algorithm items
- Protocol Reliability = $\frac{\text{Number Agree}}{\text{Total Number of Items}} \times 100$
- Algorithm Reliability = $\frac{\text{Number Agree on Algorithm}}{\text{Total Number of Algorithm Items}} \times 100$

*Do **not** include items consensus-coded as 8 or 9 in numerator or denominator, unless you coded differently.

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Protocol Reliability

- Step 1: Add coding sections to the front of the protocol.
- Step 2: After each coding section is completed, count all the items where you agreed with consensus coding and add to the front of your protocol.
- Step 3: Add all items you had in agreement between all coding sections; this will be your numerator.
- Step 4: Put your numerator over the number of items in the protocol (example: $17/34 = 50\%$).

ADOS-2 Pre-Verbal/Single Words
Age Recommendation:
31 Months and Older

Example

A Codes: 5
B Codes: 4
C Codes: 1
D Codes: 4
E Codes: 3

17 correct
34 Total Items

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Algorithm Reliability

ADOS-2 Module 1 Algorithms

Child ID: _____ Examiner: _____
Gender: Female Male Date of Birth: _____ Date of Evaluation: _____ Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES

• Convert assigned ratings of 3 to algorithm scores of 2
• Convert assigned ratings other than 0, 1, 2, or 3 (e.g., 2.5, 4, 5, and 6) to algorithm scores of 2
• Transfer assigned ratings of 0, 1, and 2 directly to the algorithm form (do not change)

Social Affect (SA)

Communication
Frequency of Spontaneous Vocalization Directed to Others (C-1)
Phrasing (C-2)
Gestures (C-3)
Reciprocal Social Interaction
Unusual Eye Contact (C-4)
Facial Expression (C-5)
Integration of Gaze (C-6)
Shared Enjoyment in Play (C-7)
Showing (C-8)
Spontaneous Initiation of Joint Attention (C-9)
Response to Joint Attention (C-10)
Quality of Social Overtures (C-11)

Restricted and Repetitive Behavior (RRB)

Restricted and Repetitive Behaviors
Intention of Vocalizations or Verbalizations (A-1)
Stereotyped/Idiosyncratic Use of Words or Phrases (A-2)
Unusual Sensory Interest in Play Material/Persons (B-1)
Hand and Finger and Other Complex Motor Patterns (B-2)
Unusually Repetitive Interests or Stereotyped Behaviors (B-3)

Example

Make a column next to your codes and write in trainer's codes.

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Algorithm Reliability (cont.)

ADOS-2 Module 1 Algorithms

Child ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____ Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES
 • Convert assigned ratings of 3 to algorithm scores of 2.
 • Convert assigned ratings other than 1, 2, or 3 (i.e., 7, 8, and 9) to algorithm scores of 0.
 • Transfer assigned ratings of 0, 1, and 2 directly to the algorithm form (do not convert).

FEW TO NO WORDS
 Assigned rating of 1 or 4 from AD, "Overall Level of Non-Verbal (Spoken) Language"

SOME WORDS
 Assigned rating of 0, 1, or 2 from AD, "Overall Level of Non-Verbal (Spoken) Language"

Social Affect (SA)
 Communication
 Frequency of Spontaneous Vocalization Directed to Others (A-7) _____
 Pointing (A-7) _____
 Gestures (A-8) _____
 Reciprocal Social Interaction
 Unusual Eye Contact (B-3) _____
 Facial Expressions Directed to Others (B-3) _____
 Integration of Gaze and Other Social Overlures (B-4) _____
 Shared Enjoyment in Interactions (B-5) _____
 Showing (B-9) _____
 Spontaneous Initiation of Joint Attention (B-10) _____
 Response to Joint Attention (B-11) _____
 Quality of Social Overlures (B-12) _____
SA TOTAL _____

Restricted and Repetitive Behavior (RRB)
 Restricted and Repetitive Behaviors
 Intonation of Vocalizations or Verbalizations (A-3) _____
 Stereotyped/Idiosyncratic Use of Words or Phrases (A-5) _____
 Unusual Sensory Interest in Play Material/Person (D-1) _____
 Hand and Finger and Other Complex Motor Stereotypes (D-2) _____
 Unusually Repetitive Interests or Stereotyped Behaviors (D-4) _____
RRB TOTAL _____
OVERALL TOTAL (SA + RRB) _____

Circle your codes that are the same as the trainer's codes.

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Algorithm Reliability (cont.)

ADOS-2 Module 1 Algorithms

Child ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____ Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES
 • Convert assigned ratings of 3 to algorithm scores of 2.
 • Convert assigned ratings other than 1, 2, or 3 (i.e., 7, 8, and 9) to algorithm scores of 0.
 • Transfer assigned ratings of 0, 1, and 2 directly to the algorithm form (do not convert).

FEW TO NO WORDS
 Assigned rating of 1 or 4 from AD, "Overall Level of Non-Verbal (Spoken) Language"

SOME WORDS
 Assigned rating of 0, 1, or 2 from AD, "Overall Level of Non-Verbal (Spoken) Language"

Social Affect (SA)
 Communication
 Frequency of Spontaneous Vocalization Directed to Others (A-7) _____
 Pointing (A-7) _____
 Gestures (A-8) _____
 Reciprocal Social Interaction
 Unusual Eye Contact (B-3) _____
 Facial Expressions Directed to Others (B-3) _____
 Integration of Gaze and Other Social Overlures (B-4) _____
 Shared Enjoyment in Interactions (B-5) _____
 Showing (B-9) _____
 Spontaneous Initiation of Joint Attention (B-10) _____
 Response to Joint Attention (B-11) _____
 Quality of Social Overlures (B-12) _____
SA TOTAL _____

Restricted and Repetitive Behavior (RRB)
 Restricted and Repetitive Behaviors
 Intonation of Vocalizations or Verbalizations (A-3) _____
 Stereotyped/Idiosyncratic Use of Words or Phrases (A-5) _____
 Unusual Sensory Interest in Play Material/Person (D-1) _____
 Hand and Finger and Other Complex Motor Stereotypes (D-2) _____
 Unusually Repetitive Interests or Stereotyped Behaviors (D-4) _____
RRB TOTAL _____
OVERALL TOTAL (SA + RRB) _____

Step 1:
Count up the number of codes you got correct.

Example = 6 correct

Step 2:
Divide the number of correct items by 14 to obtain percent correct/reliability.

(Ex: 6/14 = 53%)

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Summary: ADOS-2 Administration and Scoring

Common ADOS-2 Errors: Administration

- Not reading the Manual carefully; just relying on the protocol in carrying out each task
- Not getting enough practice
- Poor pacing
 - Taking too long to administer or rushing and not taking long enough
- Non-standard administration
 - Giving the wrong module
 - Using non-standardized materials
 - Skipping questions or tasks you don't think apply

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Common ADOS-2 Errors: Administration (cont.)



- Errors in prompting
 - Too much or too little
 - Not following the specified hierarchy
- Not taking notes; relying on observer to take the notes and/or score
- Not scoring immediately after the administration
- No caregiver present for Modules T-2

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Common ADOS-2 Errors: Coding

- Coding everything as 2, because the child is “clearly autistic”
- Under-coding, because the child is doing well “for someone with autism”
- Neglecting to *read each code and description carefully*
- Coding based on “impressions” rather than observable behaviors
- Not converting 3 to 2 (or 7, 8, 9 to 0) on the algorithm

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Interpretation of ADOS-2 Scores



ADOS-2 Is a Diagnostic *Instrument*

- Autism and autism spectrum disorder (ASD) cutoffs were selected to balance sensitivity and specificity of classifications in comparison to overall clinical diagnosis.
- Receiving a classification of autism or ASD does NOT necessarily mean that a clinical diagnosis of ASD is warranted.
 - Sensitivity and specificity of classifications may vary depending on the population in which it is employed.
 - Responsible use involves knowing how your patient population compares to the validation samples.
- In the age of DSM-5, the ASD vs. autism classification is not as significant; different research or clinical groups may choose to use one over the other to maximize sensitivity or specificity.

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Measuring Accuracy of Classifications: Sensitivity and Specificity

- **SENSITIVITY** = How many children with XX condition are accurately classified as having XX on the diagnostic measure
 - Sensitivity of ADOS-2: How many children **with a clinical diagnosis of ASD** score in either the ASD or autism range
- **SPECIFICITY** = How many children who do NOT have XX condition are accurately classified as NOT having XX on the diagnostic measure
 - Specificity of ADOS-2: How many children **without a clinical diagnosis of ASD** score in the non-spectrum range

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ADOS-2 Accuracy (i.e., Agreement With Overall Clinical Diagnosis)

- Updated validity sample included:
 - $n = 2724$ cases with ASD
 - $n = 881$ cases with non-spectrum disorders or typical development (including individuals diagnosed with intellectual disability, language disorders, ODD, ADD and/or ADHD, genetic syndromes, mood and anxiety disorders)
- Drawn from ASD specialty clinics
 - Children referred for clinical concerns about ASD
 - Recruited as cases or non-ASD controls for ASD-focused research studies

Source: Bishop & Lord, 2016, personal communication.

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Module 1: Sensitivity and Specificity

Among Individuals With a Nonverbal Mental Age of at Least 15 Months

	ASD (n=966)	Non-ASD (n=170)
Male (%)	82%	61%
Mean Age (SD)	4.8 Years (3.2)	3.3 Years (2.25)
Mean Verbal IQ (SD)*	37 (20)	58 (24)
Mean Nonverbal IQ (SD)*	60 (22)	69 (27)
Non-ASD Diagnosis		
Typical/No Diagnosis (%)		26%
Language Disorders (%)		19%
Intellectual Disability (ID) (%)		19%
Down Syndrome (%)		16%
Other/Missing (%)		19%
Module 1	Sensitivity	Specificity
Autism Cutoffs	.92	.85
ASD Cutoffs	.98	.74

*Does not include TD

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Module 1: Sensitivity and Specificity (cont.)

Among Individuals With a Nonverbal Mental Age of LESS Than 15 Months

	ASD (n=90)	Non-ASD (n=26)
Male (%)	79%	42%
Mean Age (SD)	3.4 Years (1.9)	2.1 Years (1.2)
Mean Verbal IQ (SD)*	21 (14)	28 (15)
Mean Nonverbal IQ (SD)*	35 (17)	37 (23)
Non-ASD Diagnosis		
Typical/No Diagnosis (%)		35%
Intellectual Disability (ID) (%)		38%
Other (%)		12%
Missing (%)		15%
Module 1	Sensitivity	Specificity
Autism Cutoffs	1.0	.50
ASD Cutoffs	1.0	.34

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Module 2: Sensitivity and Specificity

	ASD (n=543)	Non-ASD (n=213)
Male (%)	80%	73%
Mean Age (SD)	5.8 Years (3)	5 Years (3.3)
Mean Verbal IQ (SD)	73.3 (24.5)	81.9 (22)
Mean Nonverbal IQ (SD)	83.7 (23.8)	86.5 (24.6)
Non-ASD Diagnosis		
Language Disorder (%)		42.4%
Intellectual Disability (%)		17.7%
Typical/No Diagnosis (%)		13.1%
Other (%)		26.7%
Module 2	Sensitivity	Specificity
Autism Cutoffs	.85	.86
ASD Cutoffs	.94	.69

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Module 3: Sensitivity and Specificity

	ASD (n=725)	Non-ASD (n=306)
Male (%)	83%	68%
Mean Age (SD)	9 Years (3)	8.7 Years (2.9)
Mean Verbal IQ (SD)	96.6 (21)	95.7 (21.3)
Mean Nonverbal IQ (SD)	98.7 (20.1)	93.8 (21.3)
Non-ASD Diagnosis		
ADHD (%)		38.3%
Mood/Anxiety (%)		16.8%
Language Disorder (%)		13.2%
Other (%)		31.8%
Module 3	Sensitivity	Specificity
Autism Cutoffs	.79	.86
ASD Cutoffs	.92	.76

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Module 4: Sensitivity and Specificity of Revised Module 4 Algorithm (Hus and Lord, 2014)

	ASD (n=198)	Non-ASD (n=77)
Male (%)	80%	71%
Mean Age (SD)	20.3 Years (5.8)	22 Years (8.5)
Mean Verbal IQ (SD)	100.6 (21.9)	98 (23.2)
Mean Nonverbal IQ (SD)	98 (22.2)	98 (24.7)
Non-ASD Diagnosis		
Mood/Anxiety (%)		29.6%
Typical/No Diagnosis (%)		26.8%
Intellectual Disability (%)		16.9%
Other (%)		26.7%
Module 4	Sensitivity	Specificity
ASD Cutoffs	.90	.75

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Important Caveats About the Sensitivities and Specificities of the ADOS-2

- The ADOS-2 sensitivity and specificity values are based on data from examiners who are *research reliable*.
- Sensitivity and specificity of the ADOS-2 is likely to be lower when used by examiners who are not research reliable.
- All examiners (research reliable or not) must take care to interpret the ADOS-2 results in the context of all other information.
 - Examiners should also consider differences between their clinical population and the validation samples in terms of medical complexity, cultural diversity, behavioral issues, etc.

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ADOS-2 Comparison Scores

Purpose:

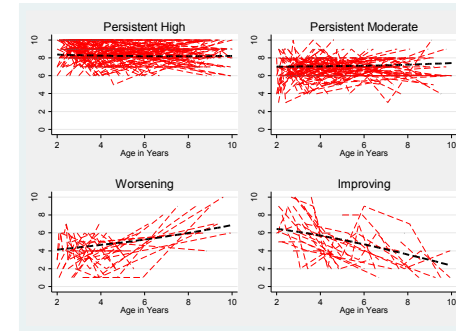
- To indicate level of autism-related symptomatology, as observed during the ADOS-2, as compared with children who have ASD and are of the same chronological age and language level
- To help interpret an individual's progress over time (i.e., across age and modules)

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Trajectory Classes

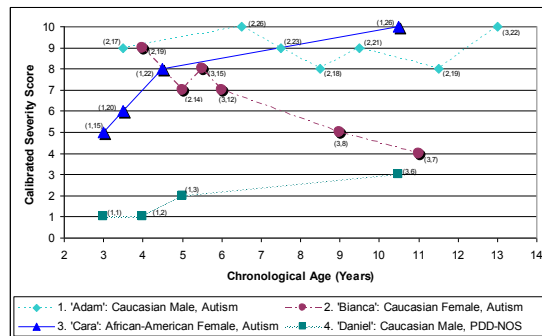
Modeled With Risk Variables (VIQ, NVIQ, Gender, Race)



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Case Summaries of Longitudinal Calibrated Severity Scores



Source: Gotham, Pickles, & Lord, 2009.

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Domain-Calibrated Scores

Table 2
Mapping of ADOS raw domain totals onto calibrated severity scores

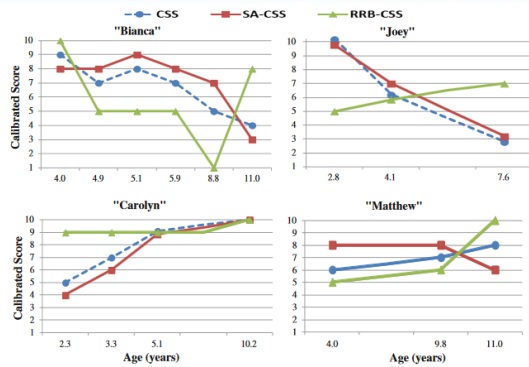
Domain	Calibrated severity scores	Raw domain totals									
		Module 1: no words		Module 1: some words		Module 2		Module 3			
		2 years	3 years	4-14 years	2-3 years	4 years	5-14 years	2-3 years	4 years	5-14 years	6-16 years
Social affect domain	1	0-3	0-3	0-2	0-1	0-1	0-1	0-1	0-1	0-1	0-1
	2	4-5	4-5	3-5	2-4	2-3	2-3	2	2-3	2	3
	3	6-8	6-9	6-9	5	4-5	4-5	4	3-4	4-5	3-4
	4	9	10	10	6-7	6-7	6-7	5	5-6	6	5
	5	10-13	11-12	11-12	8	8-9	8-9	6	7	7	6-7
	6	14-16	13-16	13-14	9-11	10-12	10-13	7-8	8-9	8-9	7-8
	7	17	17	15-16	12-13	13	14-15	9-10	10-11	11-13	9-10
	8	18	18	17-18	14-15	14-15	16	11	12-13	12-15	11-12
	9	19	19	19	16-17	16-17	17-18	12-14	14-15	16	16-17
	10	20	20	20	18-20	18-20	19-20	15-20	16-20	17-20	15-20
Restricted and Repetitive behavior domain	1	0	0	0	0	0	0	0	0	0	0
	2										
	3										
	4										
	5	1	1	1-2	1	1	1	1	1	1	1
	6	2	2-3	3	2	2	2-3	2	2-3	2-3	
	7	3	4	4	3	3-4	4	3	4	4	2
	8	4	5	5-6	4	5	5	4	5	5	3
	9	5	6	7	5	6	6	5-6	6	6	4
	10	6-8	7-8	8	6-8	7-8	7-8	7-8	7-8	7-8	6-8

Source: Hus, V., Gotham, K., & Lord, C. (2012). Standardizing ADOS Domain Scores: Separating severity of social affect and restricted and repetitive behaviors. *Journal of Autism and Developmental Disorders*, 44(10), 2400-2412. doi: 10.1007/s10803-012-1719-1

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CSS for SA and RRB Domains



Source: Adapted from Hus, Gotham, & Lord, 2014.

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What the Comparison Score Is NOT

- Does NOT represent the child's ASD symptoms or overall functioning relative to a typically developing reference group
 - It is in reference to children with ASD of the same age and language level, within the ADOS-2 context
- Is NOT based on population or epidemiological sample
 - It is derived from a large clinical/research sample
- Language-based module choice, IQ scores, and adaptive behavior scores tell you more about an individual's overall level of functioning than do Comparison Scores**

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Appropriate Use of the ADOS-2



Clinician Choice

- The licensed clinician/expert diagnostician chooses the appropriate measures to give to determine a diagnosis.
- The ADOS-2 may not be an appropriate test to use in every case (see Validity Considerations).



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Clinician Choice: Helping the Family

- Paul Wellstone and Pete Domenici Mental Health Parity and Addiction Equity Act of 2008 (MHPAEA)
 - Prohibits differential financial requirements or treatment limitations on mental healthcare.
- Wit v. United Behavioral Health (2019)
 - The court found UBH used flawed medical necessity criteria.
 - UBH was ordered to reprocess nearly 67,000 previously denied claims for mental health and substance-use treatment.
 - The Wit decision represented the first time a health plan was found to deny care by relying on improper clinical guidelines to make claim decisions.

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Clinician Choice: Helping the Family (cont.)

- If an agency attempts to mandate a specific battery:
 - This cannot legally be mandated by an insurance company
 - If they do, use MHPAEA-Disclosure-Template.pdf

OMB Control No. 0938-1080
Expiration Date: 05/31/2021

**FORM TO REQUEST DOCUMENTATION FROM AN EMPLOYER-SPONSORED
HEALTH PLAN OR A GROUP OR INDIVIDUAL MARKET INSURER CONCERNING
TREATMENT LIMITATIONS**

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Validity Considerations

- The ADOS-2 is not necessarily appropriate for every person, clinical situation, or study.
 - Modules 1–3 require nonverbal mental age of at least 15–18 months.
 - Not validated for people with blindness or deafness.
 - Also not initially designed for those who are proficient AAC users.
 - Not validated for people who cannot walk independently.
 - Not always appropriate for older children/adults with very limited language.
 - Not always appropriate for people with selective mutism, extreme anxiety, severe behavioral issues, or severe developmental delay.

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Validity Considerations (cont.)

- Cultural appropriateness
 - Examiners should be mindful of the role of race, ethnicity, and gender when administering and interpreting the ADOS-2
 - Culturally appropriate pretend play (e.g., pizza party instead of birthday party)
- Must be administered by the examiner in the individual's dominant language
 - ADOS-2 has been translated into several languages
 - If the child speaks and understands some English, but is more proficient in another language, you may still be able to administer but not score to get qualitative information, depending on overall language level
 - **Use of interpretation services/translators during the ADOS-2 is NOT appropriate**



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WPS Telehealth Statement

Can I administer the ADOS-2 remotely?

The ADOS-2 requires administration by a trained examiner who is in the same room as the examinee and in close contact, interacting with the same materials. **As such, it is not possible to validly administer this assessment remotely.** Administrations of the ADOS-2 should be delayed until it is possible to resume in-person contact. Specific to Module 4, although this module is less reliant on toys and materials than the other ADOS-2 modules, it still has required tasks that utilize materials, and the focus is still on the social interactions between examiner and examinee. Remote administration would require modifications to standard procedures, and such modifications are not recommended, as they are likely to change the social “press,” will limit the scope of observation, and may not produce valid results consistent with published research. Any administrations conducted in this manner should be for qualitative purposes only, and algorithms should not be used.

Source: <https://pages.wpspublish.com/telepractice-101#faq>

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WPS Telehealth Statement (cont.)

During COVID-19, can I validly administer the ADOS-2 while practicing social distancing and/or wearing personal protective equipment?

The ADOS-2 is used by a variety of professionals in many different practice settings around the world. As such, any given professional using the ADOS-2 during COVID-19 should review current practice guidelines for their area and recommendations from their local department of health and credentialing agencies to determine whether a standardized ADOS-2 administration can be conducted in their setting at this time. Standardized administration of the ADOS-2 involves close contact and the use of shared materials. The focus of the assessment is on the social interaction between examiner and examinee, and maintaining an unusual physical distance for a social interaction and/or wearing a face mask/plastic face shield may change the nature of that social interaction in a variety of ways, such as by interfering with establishing rapport and obscuring facial expressions and spoken language.

Source: <https://pages.wpspublish.com/telepractice-101#faq>

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WPS Telehealth Statement (cont.)



Data supporting the validity of the ADOS-2 are all based on administrations in which examiners engage with examinees without physical barriers and with a variety of physical distance depending on the specific activities being administered. The extent to which any specific variation to standardized procedures may impact the validity of an ADOS-2 assessment is an empirical question and will likely depend on aspects of the specific administration. For example, it is likely to be easier to follow standardized administration procedures while maintaining some physical distance for administration of Module 4 than for the Toddler Module. **Any variations from standard procedures should be noted in the evaluation report. Depending on the extent of variation, it may be preferable to report qualitative observations in lieu of formal scoring.**

Source: <https://pages.wpspublish.com/telepractice-101#faq>

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Telehealth Resources

- TELE-ASD-PEDS:
 - <https://vkc.vumc.org/vkc/triad/tele-asd-peds>
 - Wagner, L., Corona, L. L., Weitlauf, A. S., et al. (2021). Use of the TELE-ASD-PEDS for autism evaluations in response to COVID-19: Preliminary outcomes and clinician acceptability. *Journal of Autism and Developmental Disorders*, 51(9), 3063–3072.
- Brief Observation of Symptoms of Autism (BOSA):
 - https://www.semel.ucla.edu/sites/default/files/autism/pdf/BOSA_Training_FAQ
 - Dow, D., Holbrook, A., Toolan, C., McDonald, N., Sterrett, K., Rosen, N., Kim, S., & Lord, C. (2021). The Brief Observation of Symptoms of Autism (BOSA): Development of a new adapted assessment measure for remote telehealth administration through COVID-19 and beyond. *Journal of Autism and Developmental Disorders*, 52(12), 5383–5394.
- Childhood Autism Rating Scale (CARS):
 - <https://www.wpspublish.com/cars-2-childhood-autism-rating-scale-second-edition>
 - Schopler, E., Reichler, R. J., & Renner, B. R. (2010). The childhood autism rating scale (CARS). Western Psychological Services.
- Naturalistic Observation Diagnostic Assessment (NODA):
 - <https://www.nodaautismdiagnosis.com/>
 - Nazreen, N., Rozga, A., Smith, C. J., Oberleitner, R., Abowd, G. D., & Arriaga, R. I. (2015). A novel system for supporting autism diagnosis using home videos: Iterative development and evaluation of system design. *JMIR mHealth and uHealth*, 3(2), e4393.

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A Useful Tool but Not the Only Tool

- The ADOS-2 will **NOT** provide you with every answer you seek
- Additional components are necessary to answer various clinical and research questions relevant to ASD, such as:
 - Developmental/cognitive testing
 - Measures of comorbid symptoms
 - Medical history
 - Developmental history
 - School functioning
- Provides framework for identifying areas of social and communication difficulty relevant for informing your diagnostic decisions (and/or treatment planning)

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Incorporating ADOS-2 Results Into Evaluation Reports

- Provide description of ADOS-2 (why administered, which module, types of activities)
- Report diagnostically relevant behaviors observed during the ADOS-2 (social–communication and repetitive behaviors)
- Do not report raw scores
- If validity criteria are met and it is otherwise appropriate to do so, may report classification and/or comparison scores

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ADOS-2 Summary of Findings Example

- The ADOS-2 results in a cutoff score indicating whether a pattern of behaviors is consistent with a classification of autism or autism spectrum, or not consistent with ASD (“non-spectrum”). On this administration of the ADOS-2 Module 1, [name’s] scores exceeded cutoffs for an ADOS-2 classification of autism.
- The ADOS-2 comparison score measures an individual’s level of ASD-related behaviors compared to other individuals with ASD of the same age and expressive language level. On this administration of the ADOS-2 Module 1, [name] received a Comparison Score of 8, indicating a high level of ASD-related behaviors exhibited during the ADOS-2.

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Was my ADOS valid?

- Things you can more or less control:
 - Standardized administration and materials
 - Caregiver present for T/1/2
 - Adequate time (at least 30 minutes but no more than 60 minutes)
- Things you cannot necessarily control:
 - Child behavior, emotional state during ADOS
 - Child developmental level
 - Child motor skills, vision, hearing
 - COVID-19 protocols for use of PPE, etc.

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What if you feel it is not valid?

- An attempt was made to administer the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) Module T to [name]; however, all criteria for ADOS-2 validity were not met, specifically in the area of [insert validity concern]. Instead, qualitative observations can be made from several play-based tasks that were administered.
- (Proceed describing what behaviors were observed related to language, communication, reciprocal social interaction, stereotyped behaviors, play skills, and E codes.)

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ADOS-2 Description Module 1

- As part of this evaluation, [child's name] was given the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) Module 1, which is designed for children who are pre-verbal or use single words to communicate. The ADOS-2 is a structured observation designed to elicit social and communication behaviors in children suspected of having ASD. Module 1 involves structured and unstructured tasks, during which the examiner engages in a variety of interactions with the child. Module 1 includes opportunities for adult-led interactions, such as having a pretend birthday party for a doll, playing with bubbles and balloons, and imitating actions with objects, as well as opportunities to observe the child in spontaneous play during free play and snack activities.

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ADOS-2 Social Affect Summary Young Child

- In the area of social communication, Eve did not consistently use eye contact. She used word approximations, but she did not consistently direct these vocalizations to others. More frequently, she took the examiner's hand and placed it on objects of interest. In one instance, Eve used an approximation of a point to request a snack. When the examiner pointed to a toy bunny, Eve responded by turning her head and saying, "bu." She also responded by making eye contact and smiling when her mother called her name. Eve seemed to enjoy many of the ADOS-2 activities. She did not generally share her enjoyment with others, though she did laugh briefly when the examiner tickled her.

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ADOS-2 RRB Summary Young Child

- John enjoyed several of the ADOS-2 activities, including make-believe play and the pretend birthday party. During make-believe play with a family of dolls, John placed the dolls on the toy sofas and had a doll carry miniature food. While having a pretend birthday party for a doll, John participated by singing along. He particularly liked it when the examiner pretended to spill the juice during the party, so he imitated this action several times. John also tended to play with toys or objects in a repetitive way and to focus on the sensory aspects of different objects. For example, when using a bubble blower, John visually inspected the spinning fan and peered at it out of the corner of his eye.

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ADOS-2 Description Module 3

- As part of this evaluation, [child's name] was given the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) Module 3, which is designed for children who speak in full sentences. The ADOS-2 is a structured observation designed to elicit social and communication behaviors in children suspected of having ASD. Module 3 involves structured and unstructured tasks, during which the examiner engages in a variety of interactions with the child. Module 3 includes opportunities for conversation, make-believe play, telling a story from a book, describing a picture, and answering questions about emotions and relationships. The ADOS-2 results in a cutoff score indicating a pattern of behaviors consistent with autism, consistent with a milder classification of autism spectrum, or not consistent with ASD ("non-spectrum").

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ADOS-2 Social Affect Summary Young Adult

- During the ADOS-2, Adam used complex sentences to communicate. When he spoke, he made grammatical errors (e.g., "Yes, I changes that!"). He did not use conventional or descriptive gestures. Conversationally, Adam engaged in back-and-forth exchanges across a variety of topics. At the same time, his responses were often tangential, and on occasion he was difficult to interrupt. When interacting, Adam made little eye contact and it was difficult to know what he was thinking or feeling based on his facial expression. With regard to his emotions, Adam identified experiences that made him feel happy, afraid, mad, and sad but he did not describe how these emotions felt. Similarly, Adam demonstrated limited insight into relationships and responsibility. For example, when asked about the pros and cons of marriage, Adam noted that a nice thing about being married would be to live with "a cool person, like a celebrity."

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ADOS-2 RRB Summary Adolescent

- Sarah enjoyed talking about comic books and movies. She demonstrated an unusually intense interest in talking about Spider-Man, often making tangential references to the character. In addition, Sarah repeatedly discussed the dangers of gang activity and other "illicit" activities. During the ADOS-2, Sarah also demonstrated compulsive behavior. For example, when talking about a recent vacation, Sarah insisted on listing all the places she has visited in alphabetical order. When the examiner tried to comment about a particular city, Sarah spoke over the examiner in order to continue listing off her places of travel in this order.

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Summary and Next Steps

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Value of Assessment = More Than Just a Diagnosis

- Accurate diagnosis is needed to access appropriate intervention services
- Assessment is needed to
 - Establish developmental and language levels
 - Determine patterns of strengths and weaknesses
 - Develop individualized behavioral and educational intervention programs
 - Educate and empower parents and teachers

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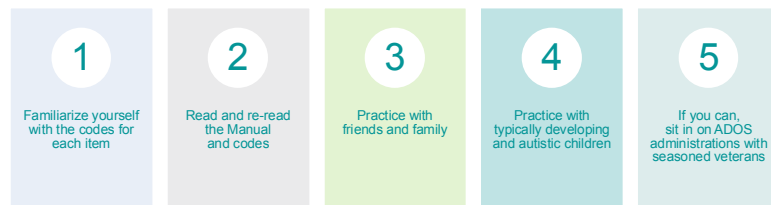
Use of the ADOS-2 in Diagnostic Assessment

- ADOS-2 creates contexts in which to observe ASD-related symptoms and behaviors
- Scores yield a classification, NOT a diagnosis
- As with all measures, some children will be incorrectly classified
- ADOS-2 should only be used by people who frequently assess individuals with ASD
- IF you choose to incorporate the ADOS-2 into your battery, additional preparation and practice will be necessary before you can use it for clinical or research purposes

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Next Steps for Perfecting Your Administration



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Tips for Mastering Administration

To be ready to use the ADOS-2 in real life:

- Be able to give the ADOS fluidly without referring to the manual
- Read through the codes and tasks right before you start administering
- It can be helpful to write down prompt hierarchy
 - (e.g., 3-2-1 or 4-2-2-1)
 - ...but still remember that you need to be available to respond to the participant's bids. Don't write so much that you're basically reading from your protocol.

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ADOS-2 Toddler Module

The ADOS-2 Toddler Module is a diagnostic tool that can be used as part of a comprehensive assessment for autism spectrum disorder (ASD) in children 12 to 30 months of age who are not yet speaking in phrases. The 1-day workshop uses lecture format and video demonstrations to introduce the basic principles of administration and scoring, providing an essential step toward competence in using the ADOS-2 Toddler Module for clinical and/or research purposes.

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Further Practice: GAIN! Global ADOS-2 Intervention Network

- GAIN is a professional development program designed to improve proficiency on the ADOS-2. GAIN is a virtual, live-training webinar hosted by ADOS-2 trainers.
- GAIN meets the fourth Friday of each month, and is divided into quarters, each focused on a different module of the ADOS-2.
- Additional sessions have been added to accommodate different time zones.
- Join GAIN here: <https://psychiatry.weill.cornell.edu/education-training/autism/gain>
- Email the GAIN training coordinator at GAIN@med.cornell.edu with any questions.

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After the Course

- Access to post-course videos for Modules 1–4 and completed protocols for further viewing
- Email questions about post-course videos and/or other lingering questions
- Completion certificates will be available in WPS ProLearn

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Contact Information

- WPS
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- Designated training sites for the ADOS-2 and ADI-R:
 - UCSF Center for ASD and NDDs
<https://autism.ucsf.edu/training>
 - Center for Autism and the Developing Brain (CADB)
<https://psychiatry.weill.cornell.edu/education-training/autism-training>
 - University of Minnesota (UMN)
<https://med.umn.edu/pediatrics/divisions/clinical-behavioral-neuroscience/education/autism-training>
 - Rutgers University
<https://rcaas.rutgers.edu/about/professional-training>
 - SpectrumKU (Korea University)
<https://spectrumku.com/main/main.html>

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ADOS-2 Additional Resources and Explainers

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Different Types of Gestures (as Defined by the ADOS-2)

REMEMBER: To be considered a gesture, it must be communicative; in general, it is not considered a gesture if the participant is holding something in their hand.

- **Descriptive:** pantomime actions used when trying to describe or represent an action, object, or event for communicative purposes (e.g., size, fishing cast, flying, blow to request blowing a balloon, brushing teeth)
- **Conventional:** have a standard social meaning within a particular culture; learned or modeled (e.g., thumbs up, wave, high five, clap, nodding/shaking head)
- **Instrumental:** intended to achieve a goal (e.g., beckoning someone to come here, communicative reach)

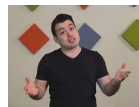


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Different Types of Gestures (as Defined by the ADOS-2) (cont.)

- **Informational:** to convey specific information (e.g., holding up two fingers to mean "two")
- **Emotional:** express an emotion (e.g., shaking fists to mean angry, putting one's hand to one's mouth and gasping to indicate shock)
- **Emphatic:** hand movements (beats) that are integrated into speech to provide emphasis as one is talking (Module 4 only)



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Facial Expressions

- Look for:
 - Range
 - Directedness
- Can be subtle, especially with older participants
- Need to be directed toward people
 - When eye contact is already low, look for other signs of directedness

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Social Overtures vs. Social Responses

Social Overtures	Social Responses
- Behavior <u>initiated</u> by the examinee - Any attempt to get or maintain the examiner's attention for any purpose	- Behavior <u>in response</u> to another person OR to the situation - Absence of an expected behavior (such as ignoring examiner) - Consider: - Response to presses - Response to social context - Absence of appropriate responses

Remember: Consider both quality and frequency

(see ADOS-2 Manual, p. 19)

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Behaviors to Consider in Social Overtures and Responses

	Social Overtures	Social Responses	Social Overture or Social Response
Verbal	- Offering information during a conversation after a pause - Asking questions - Greeting - Requesting - Commenting - Asking for clarification	- Answering questions - Refusing/protesting - Making non-lexical conversational sounds - Making undirected vocalizations	- Intonation - Stereotyped speech - Bringing up preoccupation - Seeking/offering comfort - Sharing enjoyment (verbal) - Screaming/yelling
Nonverbal	- Initiating by showing a toy or object - Giving a toy or object without prompt - Using spontaneous joint attention to direct attention - Moving closer to indicate enjoyment/continuation - Requesting (nonverbal) - Sharing without prompt - Initiating a game	- Nodding as someone talks - Modulating eye contact - Changing facial expressions - Moving away/withdrawing - Turning to name - Responsive social smile - Reciprocating in a game	- Eye contact - Being overly aggressive or impulsive - Changing facial expressions - Joining an activity (physically) - Pointing/gesturing - Moving another's hand or body - Odd affect/laughing

Could be either (or neither): Context determines whether the behavior is an overture or response

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Amount of Reciprocal Social Communication

Social Overtures	Social Responses	Amount of Reciprocal Social Communication
- Behavior <u>initiated</u> by the examinee - Has communicative intent - Attempt to get or maintain person's attention	- Behavior <u>in response</u> to another person or the situation - Absence of an expected behavior (such as ignoring examiner) - Consider: - Response to presses - Response to social context - Absence of appropriate responses	- Behaviors that are in the service of the social interaction - Can be verbal or nonverbal - Can be very brief - Can be sophisticated or "low level" - E.g., saying "thank you" or "bless you," "mm hmm" and/or nodding along with what you are saying, checking in with eye contact, trying to help when you look like you need it - In Module 2, can be directed to examiner or parent

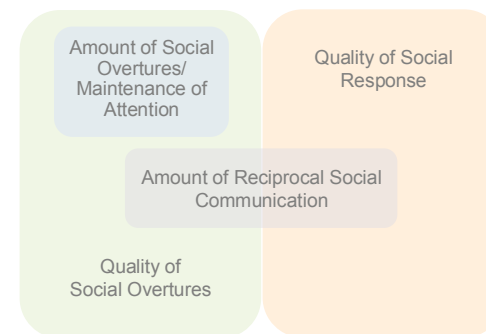
If any of these
• lead to
• or are for the
benefit of
the social interaction



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How They Fit Together



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Module 1: Code Explainer

Integration of Gaze and Other Behaviors During Social Overtures

Three Different Components

The diagram consists of three large, horizontally aligned ovals. The left oval is light blue and labeled 'Eye contact'. The middle oval is light green and labeled 'Gestures'. The right oval is light orange and labeled 'Vocalizations'.

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Integration of Gaze and Other Behaviors During Social Overtures

Code of 0 =

The diagram shows three separate ovals, each with a label inside: 'Eye contact' (light blue), 'Gestures' (light green), and 'Vocalizations' (light orange). They are arranged horizontally and separated by the word 'or'.

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Integration of Gaze and Other Behaviors During Social Overtures (cont.)

Code of 1 =

The diagram shows two separate ovals, each with a label inside: 'Eye contact' (light blue) and 'Gestures' (light green). They are arranged horizontally and separated by the word 'and'. To the left of these ovals is a vertical bracket labeled 'independent'. To the right of this pair is the word 'or', followed by another pair of ovals: 'Eye contact' (light blue) and 'Vocalizations' (light orange), also separated by the word 'and'.

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Integration of Gaze and Other Behaviors During Social Overtures (cont.)

Code of 2 =

The diagram shows three separate ovals, each with a label inside: 'Eye contact' (light blue), 'Gestures' (light green), and 'Vocalizations' (light orange). They are arranged horizontally and separated by the word 'or'.

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Integration of Gaze and Other Behaviors During Social Overtures *(cont.)*



Code of 3 =

Nothing

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Questions?



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